

138.6 Gb/s Net-Data Rate sub-THz Transmission over 5.5 km Enabled by Resonant Plasmonic Modulator

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Abstract A plasmonic-assisted optical-sub-THz-optical transmission link over a distance of 5.5 km, with net-data rates of 138.6 Gbps at a carrier frequency of 273 GHz is demonstrated. Only linear DSP is used to show a 762.3 Gbps km distance-net-rate product under summer conditions. ©2026 The Authors

Introduction

The exponential growth of mobile traffic, driven by streaming and future applications like self driving cars, is pushing wireless networks towards their capacity limits [1]. There is a trend towards higher carrier frequencies to increase the overall capacity. Of particular interest are the non-congested frequencies around 252–325 GHz as e.g. targeted by the IEEE 802.15.3d standard [2]. This frequency window would be attractive for switched point-to-point communications, with channel bandwidths reaching tens of GHz and potential data rates up to and beyond 100 Gbps.

Recently, high-capacity or multi-kilometer links with carrier frequencies beyond 200 GHz, have been reported using various transmitter (Tx) and receiver (Rx) technologies. Most demonstrations use photonic-assisted Tx based on uni-traveling-carrier photodiodes (UTC-PDs) and electronic mixers at Rx. With this technology, the highest achieved net-data rate (NDR) under real-world conditions is 204 Gbps over 214 m [3]. Also, a long-distance transmission over 4.6 km at an NDR of 1.7 Gbps has already been shown [4]. Alternatively, traveling-wave tube amplifiers have been employed to increase the transmitted power. This way 2.2 km have been bridged at an NDR of 27.8 Gbps [5]. Non-photonic-assisted Tx solutions employing electronic mixers both at the Tx and Rx have shown net-data rates of 59.8 Gbps over 850 m [6]. All of the above solutions depend on electronic mixers at the Rx. And while they have shown impressive results by taking advantage of the sensitive electronic circuits, they are bandwidth-limited. When targeting optical-sub-THz-optical links, they additionally require a modulator to convert the signals back to the optical domain.

Separately, plasmonic-assisted sub-THz receiver solutions have been demonstrated. This way, the need for an additional optical modulator is eliminated and the sub-THz link seamlessly integrates into the existing optical networks. This is

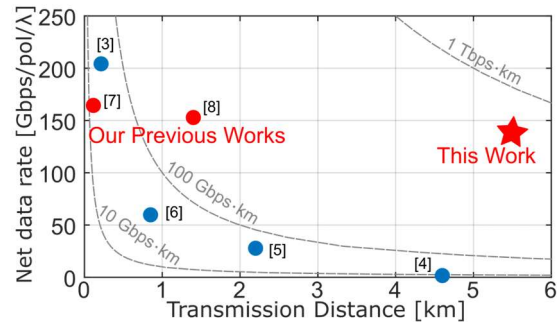


Fig. 1: Comparison of the net-data rate of recent sub-THz links above 200 GHz.

possible because the plasmonic technology overcomes the bandwidth bottleneck associated with electronic-mixer-based receivers. With this technology, NDRs of 164 Gbps over 115 m [7] and 214 Gbps over 1.4 km [8] have been achieved.

Fig. 1 shows the NDR against the transmission distances. The solid lines indicate fixed distance-net-rate products. The figure reveals the lack of high-capacity links over longer distance.

Yet, achieving longer distance link while maintaining high data rate in sub-THz free space link poses a significant challenge. The free space propagation loss (FSPL) and atmospheric absorption is considerably higher compared to conventional wireless signal frequencies, resulting in smaller received THz power as the distance increases. Additionally, bandwidth limitations and non-linear distortions in electronic receivers pose a significant bottleneck [9], [10]. Therefore, development of linear higher sensitivity, high-bandwidth receivers is crucial towards realizing higher distance-data-rate products.

In this paper, we introduce a new sensitivity-improved plasmonic sub-THz receiver (Rx). It enables a record optical-sub-THz-optical transmission over 5.5 km at a NDR of 138.6 Gbps at a carrier frequency of 273 GHz in challenging atmospheric conditions over city terrain during summer conditions.

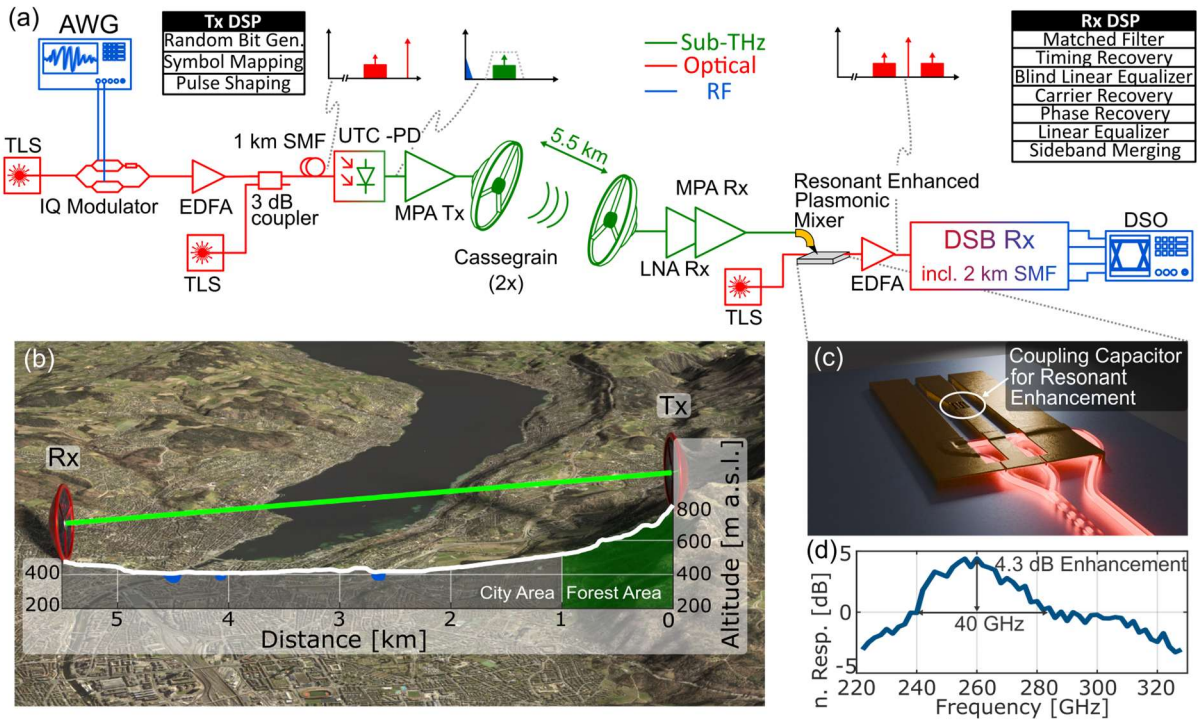


Fig. 2: (a) Depiction of the experimental setup. (b) Illustration of the air channel of the subTHz signal and overlaid the altitude above sea level along the beam path. The highest point above ground is 250 m. The crossed waterways are shown in blue. Elevation data and satellite image are based on data from ©swisstopo [12], [13] (c) Conceptual illustration of the plasmonic organic electrical resonant enhanced modulator. (d) Measured electrooptic response of the resonant enhanced modulator normalized to DC V_{π} .

The introduced resonant modulator enhancement leads to a signal enhancement of 4.3 dB. The dual-sideband receiver scheme [11], provides up to 3 dB SNR gain. In total, combining the schemes, we get an SNR gain of up to 7.3 dB. The distance-NDR product of 762.3 Gbps-km result marks – to the best of our knowledge – a new record for single-polarization single-carrier transmission at carrier frequencies above 200 GHz.

Experimental Setup

Fig. 2(a) shows the 5.5 km plasmonic-assisted Tx and optical plasmonic Rx setup. An IQ modulator was driven by an arbitrary waveform generator (AWG) with a 60 GHz bandwidth and a tunable laser source (TLS) in the C-band. The optical signal was amplified by an erbium-doped fiber amplifier (EDFA), bandpass filtered, and transmitted through a 1 km single-mode fiber (SMF). Subsequently, the signal was combined with a local oscillator (LO) laser in a 3 dB coupler. The signal and the LO were then mixed down to a sub-THz carrier at 273 GHz using a UTC-PD. The UTC-PD has a waveguide output and a DC responsivity of 0.22 A/W. This sub-THz signal was amplified with an electronic medium-power amplifier (MPA) with a gain of 24 dB. To bridge the 5.5 km free-space path, Cassegrain antennas with a gain of 62 dBi were employed at both the Tx and Rx site.

At the Rx side, after the Cassegrain antenna, the signal was first amplified by an electronic low-noise amplifier with a gain of 14 dB, followed by

an MPA providing an additional gain of 23 dB. The amplified sub-THz signal was upmixed to an optical carrier at 1550.55 nm using a plasmonic organic modulator featuring an electrical enhancement [14]. The optical signal was then amplified by an EDFA, filtered and routed through the dual-sideband receiver [15], [16], where the optical signal was converted to the radio frequency (RF) domain using four 70 GHz photodiodes (PDs) and sampled with a 110 GHz real-time oscilloscope for offline processing.

The 5.5 km sub-THz link was set up across the city of Zurich in Switzerland. The Tx was located on Uetliberg at an altitude of 808 m above sea level (a.s.l.), while the Rx was located at ETH Zurich facilities in the city center at 496 m a.s.l., resulting in an inclination of -3.24° and a maximum altitude above ground of 250 m. Around 4.5 km of the propagation path traversed a diverse urban city terrain including three waterways, while the remaining 1 km crossed forest area. See Fig. 2(b). To summarize, a link budget of the electrical path is shown in Tab. 1.

Tab. 1: Electrical Link Budget.

Tx Power	5 dBm
Antenna Gain Tx	62 dB
FSPL 5.5 km @ 275 GHz	156 dB
Estimated Atmospheric Attenuation (@ 80 GBd shot)	13 dB
Antenna Gain Rx	62 dB
LNA Gain	14 dB
MPA Gain	23 dB
Insertion Loss Probe	3 dB
Power at Modulator	-8 dBm

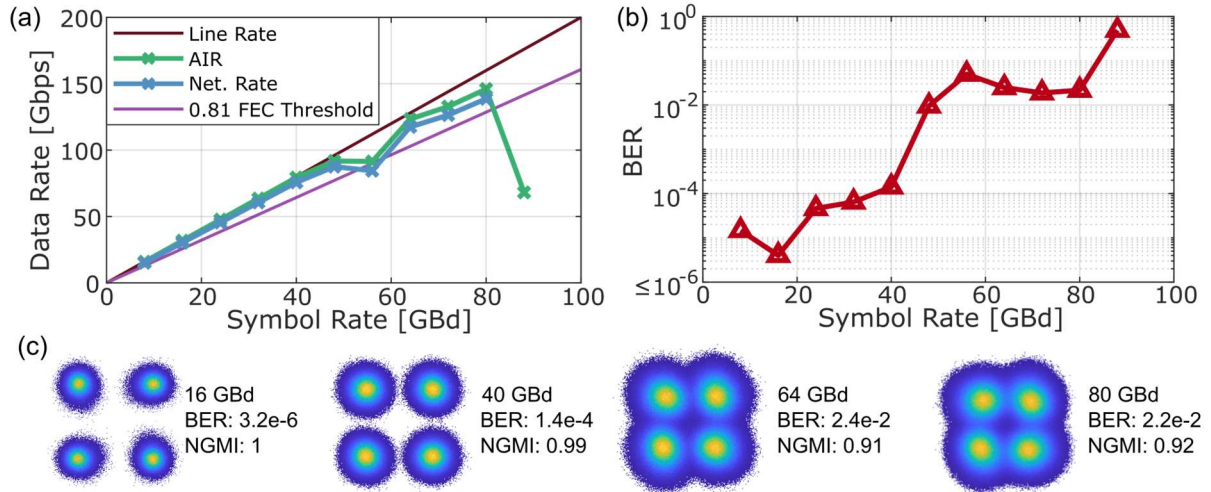


Fig. 3: Results of the transmission experiment. (a) Net data rate, achievable information rate, line rate and 0.81 FEC threshold vs. symbol rate (b) Bit error ratio vs. symbol rate. (c) Constellation diagrams recorded at symbol rates of 16, 40, 64 and 80 GBd.

Resonant Plasmonic Modulator

The plasmonic electronic-enhanced modulator, used as the mixer at the receiver, was fabricated in-house [14]. It was designed to achieve resonant enhancement at 260 GHz. A conceptual illustration of the modulator is depicted in Fig. 2(c), while the measured enhancement is shown in Fig. 2(d).

To achieve electrical enhancement, the contact circuit was designed to form a capacitively-coupled transmission line resonator [14]. At resonance, the voltage across the capacitive modulator is enhanced, improving thereby the receiver sensitivity. The modulator is edge-coupled, with a total insertion loss of 15 dB at 1550 nm. The $V_{\pi,eq}$ is reduced from 10.4 V at DC to 6.4 V at resonance, corresponding to a resonant enhancement of 4.3 dB. The resonant response provides a bandwidth of 40 GHz. See Fig. 2(d).

Data Transmission Results

Results are shown in Fig. 3. A single-carrier 4-QAM ($H = 2$) signal was sent at different symbol rates from 8 GBd to 80 GBd at a center frequency of 273 GHz. In Fig. 4 we show successful transmission at line rates of 158.75 Gbps. These translate to an achievable information rate (AIR) of 146.0 Gbps, and a NDR of 138.6 Gbps. This allows us to report a new record distance-NDR product of 762.3 Gbps·km.

These results were obtained using only linear digital signal processing (DSP) at both the Tx and the Rx. At the Tx, the DSP consisted of random bit generation, mapping the bits to the gray-coded 4-QAM symbols, and pulse shaping with a root-raised cosine filter with a roll-off of 0.05. At the Rx, the DSP consisted of a matched filter, timing recovery, a blind T/2-spaced linear equalizer with 101 taps, a 4th power carrier recovery, a pilot-based phase recovery with an overhead of 1/128 and a T/2-spaced linear equalizer with 251 taps for each of the two sidebands. Linear DSP alone

was sufficient to achieve the reported performance. Lastly, the two sidebands are combined as described in [16]. The line rate was calculated by $LR = R_s \cdot H$, where R_s is the symbol rate and H is the entropy of the transmitted QAM constellation. The achievable information rate was calculated by $AIR = R_s \cdot GMI$, where GMI is the generalized mutual information. The NDR was calculated according to $NDR = R_c \cdot LR$, where R_c is the code rate of the soft-decision forward error correction code at a normalized GMI (NGMI) threshold [17], [18]. The NGMI threshold used was 0.81, at which $R_c = 0.752$.

The measurements were conducted on a single day in July 2026, during summer weather conditions, between 16:00 and 22:30 local time. During the measurement period the absolute humidity was estimated to be between 3.2 and 4.4 g/m³ using [19], the atmospheric gaseous attenuation was estimated according to ITU-R P.676 to be between 7 and 13.6 dB.

Conclusion

In this paper, we have experimentally demonstrated a single-carrier 5.5 km sub-THz wireless link at 273 GHz in a real-world scenario. A maximum net data rate of 138.6 Gbps was achieved thanks to the electrical enhanced plasmonic modulator featuring a 4.3 dB enhancement and the SNR gain of up to 3 dB coming from the dual-sideband receiver. This results in a distance-rate product of 762.3 Gbps·km - the highest recorded in any sub-THz communication to the best of our knowledge. The wireless link was established in summer over predominantly urban terrain. These results thereby highlight the potential of the new resonant plasmonic receiver in a dual-sideband configuration for real-world wireless sub-THz links.

Acknowledgements

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